

Space Station Air Evacuation Time

Team 785, Problem A

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Abstract

Our study investigates the pressure-time relationship for a spaceship impacted by a micrometeorite, focusing on how a pressure difference evolves based on the hole diameter and wall thickness. By applying mass flow rate equations, the ideal gas law and compressible flow, we examined cases with infinitesimal wall thickness. For a wall of infinitesimal thickness, pressure equalization took approximately 16 hours, with time scaling as the square of the hole diameter. Key assumptions include adiabatic gas expansion, negligible space station velocity, uniform gas composition, and the applicability of the ideal gas law for compressible flow. These assumptions facilitate a simplified yet accurate model of pressure dynamics post-impact.

1 Introduction

1.1 Problem Restatement

In this problem, we are given a cylindrical space station with some given initial conditions that is being impacted by a micrometeorite and we are asked to find how much time is needed to achieve a given pressure change and how time depends on the diameter of the hole created by the micrometeorite.

In this article, we investigate the relationship between pressure and time for a spaceship impacted by a micrometeorite. Employing specific assumptions, we were able to define a test case in which we can restrict how the pressure difference is achieved. Utilizing equations for mass flow rate, the ideal gas law, compressible flow and our given values and assumptions, we were able to analyze the time needed to establish the pressure difference and its relationship to the diameter of the hole created by the micrometeorite. We analyzed this relationship for a test case in which the space station's walls were of infinitesimal thickness and we concluded that the time needed to establish the pressure difference was approximately 16 hours for an infinitesimal thickness, under our given conditions and assumptions. In our case of infinitesimal thickness, we found that the dependence of the time to establish the pressure difference upon the diameter of the hole created was proportional to d^2 .

2 Assumptions

The test case of a space station with infinitesimal thickness can be generated from the following assumptions. Firstly, we assume that the gas expansion when a hole is created can be considered adiabatic, an interaction where no heat is lost, and thus, temperature

can be held constant in the ideal gas law equation. We can assume that this interaction is adiabatic because there is nothing in the vacuum to lose heat to and ostensibly, nothing within the space station which the air could be reacting with to lose heat. Secondly, we can assume that the space station's velocity is negligible because in the reference frame of the spaceship, so long as it is not accelerating, its velocity will be constant and thus, in comparison to the outflow velocity which is maximally the speed of sound, the spaceship's velocity is negligible due to the conservation of momentum. We can also assume that the gas composition contained by the space station is uniform and that this gas composition does not experience a gravitational potential or that if it were to experience such a gravitational potential, the effect of this potential on the pressure is negligible as the the relatively small volume of the space station would result in very small differences in the pressure gradient. Lastly, we can assume that the ideal gas law when adjusted for compressible flow is an accurate approximation of the event's behavior as while is a simplified equation of state, it is generally accurate for diatomic gas compositions like that of air and for relatively low pressures and high temperatures and thus, our assumptions should model an accurate approximation of the necessary time to establish a pressure difference for a space station struck by a meteorite (Donev, 2024).

3 Notation

- $P(t)$: pressure inside the space station at time t .
- $P_0 = 1 \text{ atm}$, the initial pressure inside the space station.
- $P_f = 0.3 \text{ atm}$, the final pressure within the space station.
- V = Volume of the space station, it is a cylinder so $V = \pi r^2 h$.
- A = Area of the hole created by the micrometeorite.
- $T = 293K = 20C + 273$ = The temperature inside the space station.
- $\gamma = 1.4$, adiabatic index for air.
- $R = 287J/kg/K$ = specific gas constant for air.
- C_d = discharge coefficient, this accounts for non-ideal geometry of the system.

4 Theoretical Framework

In this section, we will present a justification for our conclusion through our calculations, the equations we used and the motivations for them.

4.1 Key Equations and Principles

The equations we used are the following:

Ideal Gas Law:

$$P = \frac{nRT}{V} \tag{1}$$

Mass Flow Equations (NASA, 2024a):

$$\dot{m} = C_d A P(t) \sqrt{\frac{\gamma}{RT}} \left(\frac{2}{\gamma + 1} \right)^{\frac{\gamma+1}{2\gamma-1}} \quad (2)$$

$$\dot{m} = \frac{V}{RT} \frac{dP}{dt} \quad (3)$$

Newtons Second Law:

$$\frac{dp}{dt} = \frac{dv}{dt} m \quad (4)$$

Pressure Differential Equation:

$$P = P_0 e^{-kt} \quad (5)$$

Escape Time Scaling Relation:

$$t \propto \frac{1}{r^2} \quad (6)$$

5 Test Case

We first start with a test case assuming that the ship is static. We will show that the effect of the motion of the ship is negligible, and might be used to gain a higher-order refinement to the escape time. However, from a perspective of safety of any astronauts inside the ship, it is prudent to consider an underestimation of the escape time to ensure safety in case of a similar breach.

In this initial iteration of the problem, we assume that the spaceship is at rest, that the thickness of the ship is infinitesimal, that the structural integrity of the ship is not compromised after the breach, and that the gas is ideal and the processes involved are adiabatic.

Mass loss rate:

$$\frac{dm}{dt} = -\rho \cdot A \cdot v_{out}$$

Momentum conservation:

$$\frac{d(mv)}{dt} = \frac{dm}{dt} \cdot v + m \cdot \frac{dv}{dt} = F_{net}$$

Assume the only force on the ship is due to this outflow. $F_{net} = 0$:

$$m \cdot \frac{dv}{dt} + v \cdot \frac{dm}{dt} = 0$$

Substituting $\frac{dm}{dt}$:

$$m \cdot \frac{dv}{dt} - v \cdot \rho \cdot A \cdot v_{out} = 0$$

Consider a first approximation of mass outflow

$$\frac{dm}{dt} = -\rho \cdot A \cdot v_{out}$$

Momentum conservation:

$$\frac{d(mv)}{dt} = \frac{dm}{dt} \cdot v + m \cdot \frac{dv}{dt} = F_{net}$$

Assume constant external forces $F_{net} = 0$:

$$m \cdot \frac{dv}{dt} + v \cdot \frac{dm}{dt} = 0$$

Substituting $\frac{dm}{dt}$:

$$m \cdot \frac{dv}{dt} - v \cdot \rho \cdot A \cdot v_{out} = 0$$

Resulting system of equations:

$$\frac{dv}{dt} = -\frac{\rho A}{m} v_{out}, \quad v_{out} = \sqrt{\frac{2P}{\rho}}$$

Non-negligible mass assumption:

$$\text{For small } \frac{dm}{dt}, \quad \frac{dv}{dt} \approx 0$$

Final condition:

$$\text{Initial velocity } v \text{ not greatly impacted by } \frac{dm}{dt}.$$

Now, we can set up our differential equation to solve for pressure as a function of time. Briefly, we achieve this by assuming mass as a function of time in the ideal gas law equation and then relating that to the flow equation we found for an isothermal adiabatic process in equations 2 and 3.

$$\dot{m} = C_d A P(t) \sqrt{\frac{\gamma}{RT}} \left(\frac{2}{\gamma + 1} \right)^{\frac{\gamma+1}{2\gamma-1}}$$

$$\dot{m} = \frac{V}{RT} \frac{dP}{dt}$$

Finally, when we solve the differential equation in 5, collapsing all constant terms into the factor k in the exponential argument, we achieve the result for the depressurization time from the initial pressure of 1 atm to 0.3 atm in 16.84 hours after the breach.

We stress that this is a very tentative result given that we aren't considering the thickness of the wall. However, it immediately follows that considering the thickness of the wall will not have a significant impact on the mass outflow of the air due to continuity. Furthermore, throughout the loss of air, the outflow remains choked because the critical pressure for choked flow to be established is shown here:

$$\left(\frac{P_{out}}{P_{in}} \right)_{critical} = \left(\frac{2}{\gamma + 1} \right)^{\frac{\gamma}{\gamma-1}} = \frac{2}{2.4} \approx 0.528$$

but $P_{out} = 0$ in our case, so flow is choked, so v at hole is speed of sound in air and $\dot{m}$ does not depend on downstream pressure.

6 Results

We found that under our constraints it would take 16.84 hours for the interior of the spaceship to reach a pressure of 0.3 atmospheres. We also found that our escape time scales with radius of the hole in this relationship: $t_{escape} \propto \frac{1}{r^2}$

7 Strengths & Weaknesses

In this section, we summarize some strengths and weaknesses of our results.

Some of our strengths are:

- Considering the motion of the ship.
- Differential approach to fully solving our considered system analytically,
- Well-motivating assumptions of wall thickness and choked flow.

Some of the weaknesses of our model are:

- Rotation of ship is not accounted for.
- Laminar flow is assumed.
- Pressure of external vacuum is assumed to be 0.
- Our model assumes that the pressure inside the ship is just a function of time rather than space and time and that throughout the leakage of the gas, the pressure distribution inside the space station is spatially uniform.

8 Report on the Use of AI

We used the ChatGPT-4.0 model to generate these two responses to provide us with ideas for how to approach the fluid dynamics and whether to consider friction and ultimately, didn't use the output it provided.

Query 1: hey! can you explain how i can account for friction in a choked flow? what equations can i use, and under which conditions are they valid

Output 1: Accounting for friction in choked flow involves considering how the pressure drop due to friction impacts the flow characteristics in a conduit (e.g., a nozzle, pipe, or orifice). When flow is choked, the velocity at the throat or exit reaches the speed of sound (Mach 1), and further downstream changes cannot influence upstream conditions.

Basic Overview of Choked Flow with Friction - **Choked flow** occurs when the flow rate through a duct is limited by the speed of sound at the narrowest point, and downstream conditions (e.g., pressure) do not affect the upstream mass flow rate. - **Frictional effects** can reduce the actual flow rate compared to the ideal case by dissipating energy, thus modifying the flow properties along the length of the duct.

Governing Equations 1. **Fanno Flow**: For a flow in a constant-area duct where friction is significant, we use the Fanno flow model. This model describes the behavior of a compressible flow with friction and assumes: - Adiabatic conditions (no heat transfer). - One-dimensional, steady-state flow.

2. **Key Equations for Fanno Flow**: - **Fanno Line Equation**: Relates the changes in Mach number (M), temperature (T), pressure (P), and density (ρ) due to friction along the length of the duct. - **Dimensionless Fanno Parameter**: The length-to-diameter ratio (L^*) for choked flow due to friction is given by:

$$\frac{L^*}{D} = \frac{1}{f} \left(\frac{1 - M^2}{\gamma M^2} + \frac{\gamma + 1}{2\gamma} \ln \left(\frac{(\gamma + 1)M^2}{2 + (\gamma - 1)M^2} \right) \right)$$

where: - f = Darcy-Weisbach friction factor (assumed constant), - D = hydraulic diameter, - M = local Mach number, - γ = ratio of specific heats.

3. **Conditions for Choked Flow**: - Choked flow occurs when the Mach number reaches 1 at any point in the duct. The equations above are valid for subsonic ($M < 1$) and supersonic ($M > 1$) conditions until choking occurs. - For flows starting subsonic at the inlet ($M < 1$), friction causes the Mach number to increase towards 1. Once $M = 1$ is reached, further friction causes downstream pressure and temperature changes, but the upstream mass flow rate remains constant (choked).

Step-by-Step Approach for Analyzing Choked Flow with Friction 1. **Determine initial flow properties**: At the duct inlet, specify conditions like M_1 , P_1 , T_1 , and ρ_1 . 2. **Calculate critical properties**: Use the Fanno line equations to find how the flow properties evolve as the flow approaches $M = 1$. 3. **Check the choking condition**: When $M = 1$ is reached, the flow is choked, and the maximum flow rate is given by:

$$\dot{m}_{\max} = A\rho_*a_*$$

where A is the cross-sectional area, ρ_* is the density at the choke point, and a_* is the speed of sound at that point.

4. **Solve for L** : Use the dimensionless length relation to determine the duct length L at which choking occurs.

Practical Considerations - **Isothermal vs. Adiabatic**: The Fanno flow model assumes adiabatic conditions, so for flows where heat transfer is non-negligible (e.g., long pipes with heat loss), modifications are needed. - **Friction Factor (f)**: The friction factor can vary based on whether the flow is laminar or turbulent and the roughness of the duct surface. Ensure to use an appropriate correlation (e.g., Moody chart or Colebrook equation) for f . - **Flow Development**: Ensure the flow is fully developed if the Fanno flow assumptions are applied over the length of the duct.

By considering these equations and conditions, you can analyze and account for friction's effects on choked flow in ducts, providing a more realistic model of real-world flow behavior.

Query 2: great! im also thinking about adiabatic flow and i want you to confirm my intuition. if gas is leaking out into vacuum, it's not going to have any medium outside to lose the heat to right

Output 2: Yes, your intuition is correct! In the case where gas is leaking out into a vacuum, there is no surrounding medium to conduct or convect heat away from the gas. This means that any expansion or flow occurring as the gas exits will be **adiabatic**, as there is no heat transfer between the gas and its surroundings.

In an **adiabatic process**, the energy within the gas is conserved as it expands, which results in changes in temperature and pressure. The gas can cool down as it expands due to the conversion of internal energy into kinetic energy. In a vacuum, with no medium to transfer heat to, the process remains adiabatic because there is no external environment to exchange energy with.

This assumption aligns with how space-related or high-altitude leak scenarios are modeled, where gas venting into a vacuum is typically treated as an adiabatic expansion.

References

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